

KEERTI MAGADUM

Embedded Systems | C Programming | Linux System Programming | STM32 Firmware Development | Data Structures

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PROFESSIONAL SUMMARY

Electronics and Communication Engineering graduate and Embedded Systems Engineer with hands-on experience in C programming, Embedded C, Linux system programming, Data Structures and STM32 microcontroller firmware development. Skilled in real-time application design, sensor interfacing, inter-process communication (IPC), multithreading, and communication protocols including UART, SPI, and I2C. Completed 9 project-driven embedded and software development assignments during a structured industry training program. Seeking a full-time opportunity in Embedded Systems or Software Development to apply strong problem-solving, debugging, and firmware development skills to real-world engineering challenges.

TECHNICAL SKILLS

Programming Languages: C, Embedded C, Shell Scripting, DS

System Programming: Linux Kernel System Calls, IPC (Pipe, FIFO, Shared Memory), TCP/UDP Socket Programming, Process Management (fork, exec, wait), Signal Handling

Embedded Systems: STM32 Microcontrollers, GPIO, ADC/DAC, Timers, Counters, Interrupts, Character LCD Interfacing, HAL Drivers

Communication Protocols: UART, SPI, I2C

Platforms & Tools: Linux (VS Code, Ubuntu), STM32CubeIDE, GCC, ARM-Linux-GCC, Make-files

EDUCATION

B.E., Electronics and Communication Engineering – HSIT, Nidasoshi (VTU) CGPA: 8.28	2022 – 2026
PUC (PCMB) – Govt. PU College, Majalatti 60.8%	2022
SSLC – Govt. High School, Kesti 94.56%	2020

PROFESSIONAL TRAINING

Embedded Systems Trainee – Emertxe Information Technologies, Bengaluru Jan 2026 – June 2026

- Completed structured, industry-oriented training in Embedded Systems covering C Programming, Embedded C, Linux Fundamentals, Data Structures, Microcontroller Programming, STM32 Development, and Communication Protocols.
- Built and delivered hands-on projects (listed below) spanning Linux system programming, STM32 firmware development, and embedded application development as part of the training program.

PROJECTS (EMERTXE TRAINING)

Arbitrary Precision Calculator (APC) — C, Data Structures, Doubly Linked List, Dynamic Memory Allocation

- Developed a Big Integer Calculator using doubly linked lists to represent and manipulate arbitrarily large integers, supporting addition, subtraction, multiplication, and division with high computational accuracy.
- Built modular arithmetic functions, optimized linked list operations, and strengthened expertise in dynamic memory allocation, pointer manipulation, algorithm design, debugging, and efficient data structure implementation.

Minishell – Custom Command-Line Interpreter — C, System Calls, Process Management, Pipes, Signal Handling

- Designed a UNIX-like shell supporting built-in commands, external program execution, I/O redirection, and multi-stage pipes using fork, exec, and wait system calls.
- Implemented command parsing, tokenization, and inter-process communication, gaining deep understanding of process creation and synchronization.

Lexical Analyser – Source Code Tokenizer — C Programming, String Manipulation, File handling

- Developed a lexical analyzer using deterministic finite automata (DFA) to tokenize source code into keywords, identifiers, operators, and literals, automating the first phase of compilation.
- Designed efficient token recognition and error detection/recovery logic for malformed inputs.

Fall Detection System — *STM32, Embedded C, MPU6050 Accelerometer/Gyroscope, I2C*

- Built a real-time fall detection system using accelerometer and gyroscope data to trigger alerts on abnormal orientation or acceleration changes for elderly/patient safety applications.
- Implemented I2C sensor interfacing and motion-pattern analysis algorithms in Embedded C using STM32CubeIDE.

Smart Pedometer — *STM32, Embedded C, Accelerometer Sensor, I2C, Motion Tracking*

- Developed a step-counting system using accelerometer data and step-detection algorithms for real-time fitness and activity tracking.
- Performed sensor calibration, hardware-software integration, and debugging for reliable motion analysis.

Touch Sense Control Pad — *STM32, Embedded C, Capacitive Touch Sensing, GPIO*

- Designed a button-free touch-sensitive control interface using capacitive sensing to detect touch input and trigger predefined actions.
- Configured STM32 peripherals and developed firmware for touch detection and response logic.

Temperature & Humidity Monitoring System — *STM32, Embedded C, Sensor Interfacing, UART, LCD*

- Built an embedded system to acquire, process, and display real-time temperature and humidity readings on an LCD using STM32 HAL drivers.
- Configured STM32 peripherals for continuous environmental data acquisition and monitoring.

MP3 Tag Viewer & Editor — *C, Binary File Processing, ID3v2.3, Dynamic Memory Allocation*

- Built a command-line application to read, display, and edit MP3 metadata (Title, Artist, Album, Year, Comment) using the ID3v2.3 tag format.
- Applied binary file manipulation (fread, fwrite, fseek) and dynamic memory management techniques.

Image Steganography — *C, File Handling, Bitwise Operations*

- Designed an LSB-encoding system to embed and extract secret text within BMP image files without altering visible image properties.
- Resolved bitwise manipulation challenges for reliable data embedding and retrieval.

Address Book – Contact Management System — *C, File Handling, Structures, Sorting & Searching*

- Built a console-based contact management application supporting add, edit, search, and delete operations using structured file-based data storage.
- Implemented sorting and searching algorithms for efficient data retrieval and consistency.

ACADEMIC PROJECTS

Skin Cancer Detection using Deep Learning — *Python, CNN, TensorFlow/Keras, OpenCV*

- Developed a CNN-based deep learning model to classify cancerous vs. non-cancerous skin lesions from medical images, including preprocessing and dataset balancing.

Automated Human Following Trolley — *Embedded C, Arduino, Ultrasonic & IR Sensors, Motor Control*

- Designed an autonomous trolley using sensor-based human tracking and real-time obstacle avoidance with motor driver control.